

That's John Dvorak's 2001 account of his experience with a robot that's just barely intelligent. Dvorak is no newcomer to intelligent creations—he's a world-renowned technology author and columnist. But he already sounds ancient, almost quaint. Six years on, and people are talking very differently, as you'll see.

All Too Human

Take a look at <http://snipurl.com/digrob2>. "EveR-2 Muse," designed by the Korea Institute of Industrial Technology, has been called the world's first entertainer robot. She lip-syncs, does dance moves, and such; and since people don't like seeing hideous steel structures doing these things, EveR-2 Muse has a realistic body—artificial skin, hair, a pair of you-know-whats, and, yes, artificial eyes, of which we think a better job might have been done—but who are we to complain? Still, those eyes are horrifying, and one blogger says that "that glazed, nightmarish glare speaks for itself." We conclude that you probably wouldn't feel uncomfortable with this one looking you in the eye, but what if "she" had "realistic glass eyes"?

Now, take a look at the movies at <http://snipurl.com/digrob1>. WT-5 (Waseda Talker No. 5) doesn't quite look human, but from the videos, you'll gather, even though it makes Japanese sounds, that it's alarmingly advanced in terms of speech production. The last video is the most helpful, where the bot mimics a person saying something. Where have all the nice tin-can robots gone?

Then there's professor Ishiguro, who emphasises human-like appearance in his robots: head to <http://snipurl.com/digrob3>. The page says Repliee Q1expo interacted with people "with impersonating a TV interviewer," which we're taking to mean that the bot was presented as life-like. The first video clears things up, and all of them are worth watching. If only those body movements could have been more fluid! And then you realise you're already comparing the bot to a person, which is the ultimate in flattery as far as the professor is concerned. (Chuckle, chuckle: at the bottom of the page, there's mention of Ishiguro's writhing and someone else's greed.)

For a good listing of humanoid robots, visit <http://snipurl.com/digrob4>. They can do rather more than show off by climbing stairs—and that was just a couple of years ago. Which is precisely what makes us wonder what's round the corner—and even more, in fact, about the distant future. If this were a tabloid, we'd already have been screaming, "They're coming! They're coming!"

In Our Image

Sebastian Thrun, director of the Stanford Artificial Intelligence Lab, was in the news last year for "Stanley," his robotic car. It drove itself through about 130 miles in the Mojave desert to win the DARPA Grand Challenge for self-driven cars. He makes a point about the Roomba—the most well-known robotic vacuum cleaner—that spells out something fundamental: "I think the Roomba is a mix of curiosity and cutting-edge technology and actual utility. It's not the world's best vacuum cleaner, but it's incredibly cool to have a robot vacuum cleaner."

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Coolness is certainly a big driver when it comes to consumer robotics, but why this drive towards *humanoid* robots? It could well be a multiplication thing, as in coolness multiplied by human-like equals über-cool. It's cool to have a robotic vacuum cleaner, but it would be über-cool to have an android vacuum cleaner! Still, there are more reasons we can think of.

For one, they're cheap labour. A receptionist robot costs a lot right now to hire, but when they become commonplace, it really is as simple as that—cheap labour. Then there's the obedience factor—in fact, the word "robot" comes from something in the Czech language along the lines of "doing one's bidding." Who doesn't like to order people about? And who wants his orders to be turned down with "Nah... I don't think so!"?

Robots just *don't* say "Nah... I don't think so." You tell, they do.

In terms of following orders, robots do it with *precision*. You will not see a sloppy robot—and when we get to having them as household



Some of the latest robotic creations are anything but made of tin, and sound anything but metallic. Pictured here are Repliee Q1expo (left) and EveR-2 Muse (above)

helpers, you aren't going to see water spilt on the floor after the dishes are done! It's this very precision that has given robots a bad name—a person who in all sincerity says the time is "2:52" is thought of as robot-like in a negative way. The "solution" could be to throw in some sloppiness into the robot's AI, but that's a subject of debate. Our point here is that going beyond orders being obeyed, we want them to be obeyed to perfection (in most cases!).

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Some humanoids are designed for study purposes—WT-5, which we've mentioned, was developed to study how the human vocal system works, roughly speaking.

You could think of a whole lot of other motivational factors—their entertainment value, for example. (This might soon be gone as they become more commonplace, though.) But the ultimate motivation in the design of robots that look, walk, and talk like us could well be the collective "high" of having created something in our own image. Of having played God—literally.

Looking To Japan

So make no mistake about it: we're not divining, we aren't guessing. It's all in the news. (See box *They're Crazy!*) The wave of humanoid robots has already begun.